FINITELY MANY POINTS WITH ONE-DIMENSIONAL NOETHERIAN LOCAL RINGS ON AN INTEGRAL SEPARATED SCHEME LIE IN AN AFFINE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains a commutative diagram typeset with Xy-pic; the retained chapter preamble supplies the native Xy-pic setup, so the diagram compiles exactly as in the source.

Editorial note. The counted span cites four cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- The valuative criterion of separatedness and separatedness via tensor products. Stacks Project, Schemes, Lemma schemes-lemma-separated-implies-valuative (tag 090L); Varieties, Lemma lemma-glue-separated (tag 09MW). The first excludes a valuation ring dominating two of the local rings and the second converts that into surjectivity of A_i tensor A_j -> K.
- Glueing local rings along a field and the affine containment criterion. Stacks Project, Varieties, Lemmas lemma-glue-a-bunch-of-local-rings (tag 09N1) and lemma-points-in-affine (tag 09NJ). The first builds the semi-local ring A with Spec(A_i) as an open covering and the second converts the resulting morphism into an affine open containing the points.

Source reference. Original reference 01KZ: sources/stacks-chow/tags.txt, source lines 2068–2068. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09N0: sources/stacks-chow/tags.txt, source lines 12198–12198. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09N4: sources/stacks-chow/tags.txt, source lines 12202–12202. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09NI: sources/stacks-chow/tags.txt, source lines 12216–12216. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be an integral separated scheme. Let $x_1, \ldots, x_r \in X$ be a finite set of points such that $\mathcal{O}_{X,x_i}$ is Noetherian of dimension ≤ 1. Then there exists an affine open subscheme of X containing all of $x_1, \ldots, x_r$.*

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Proof. Let K be the field of rational functions of X. Set $A_i = \mathcal{O}_{X,x_i}$. Then $A_i \subset K$ and K is the fraction field of A_i. Since X is separated, and $x_i \neq x_j$ there cannot be a valuation ring $\mathcal{O} \subset K$ dominating both A_i and A_j. Namely, considering the diagram

$$\begin{array}{ccc} \mathrm{Spec}(\mathcal{O}) & \longrightarrow & \mathrm{Spec}(A_1) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A_2) & \longrightarrow & X \end{array}$$

and applying the valuative criterion of separatedness (Schemes, Lemma 01KZ) we would get $x_i = x_j$. Thus we see by Lemma 09N0 that $A_i \otimes A_j \to K$ is surjective for all $i \neq j$. By Lemma 09N4 we see that $A = A_1 \cap \ldots \cap A_r$ is a Noetherian semi-local ring with exactly r maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$ such that $A_i = A_{\mathfrak{m}_i}$. Moreover,

$$\mathrm{Spec}(A) = \mathrm{Spec}(A_1) \cup \ldots \cup \mathrm{Spec}(A_r)$$

is an open covering and the intersection of any two pieces of this covering is $\mathrm{Spec}(K)$. Thus the given morphisms $\mathrm{Spec}(A_i) \to X$ glue to a morphism of schemes

$$\mathrm{Spec}(A) \longrightarrow X$$

mapping $\mathfrak{m}_i$ to x_i and inducing isomorphisms of local rings. Thus the result follows from Lemma 09NI. $\square$